1️⃣ Introduction

- Global supply chains are becoming increasingly complex and vulnerable to disruptions. Traditional monitoring systems rely on reported data, which is often delayed or incomplete.
- This project proposes an AI-powered system that utilizes satellite imagery and computer vision techniques to monitor logistics and industrial activities in near real-time.
- The system aims to extract measurable operational indicators from satellite images to support data-driven decision-making in logistics, investment, and supply chain management.

2️⃣ Project Idea

The project focuses on analyzing satellite images to monitor activity in:

- Seaports
- Warehouses and logistics centers
- Industrial zones
- Transportation routes and infrastructure

Using object detection and change detection techniques, the system will identify and quantify:

- Ships in ports
- Containers in yards
- Trucks around warehouses
- Industrial building expansion
- Road disruptions or congestion

The extracted information will then be transformed into structured business indicators (KPIs).

3️⃣ Project Objectives

1. Develop computer vision models to detect key objects (ships, trucks, containers, buildings).
2. Apply change detection methods to monitor activity over time.
3. Build time-series models to analyze trends and anomalies.

4. Design an early warning system for supply chain disruptions.
5. Develop an interactive dashboard to visualize trends and risk indicators.

4️⃣ Proposed KPIs (Key Performance Indicators)

- Port Congestion Index – Measures port traffic intensity and potential delays.
- Warehouse Activity Index – Estimates operational activity based on truck movements.
- Industrial Activity Score – Monitors production-related signals in industrial zones.
- Route Risk Score – Detects potential transportation risks or infrastructure disruptions.

5️⃣ Technical Methodology

Phase 1: Data Collection

- Collect satellite images (e.g., Sentinel-2 or other open-source satellite datasets).
- Select study areas (ports, industrial zones, logistics hubs).

Phase 2: Data Annotation

- Annotate objects such as ships, trucks, containers, roads, and buildings.
- Use annotation tools (e.g., CVAT, LabelImg).

Phase 3: Model Development

- Object Detection (YOLO / Faster R-CNN).
- Change Detection techniques.
- Time Series Analysis for trend monitoring.

Phase 4: KPI Generation

- Convert detection outputs into quantitative indices.
- Develop scoring algorithms to measure activity levels.

Phase 5: Visualization & Dashboard

- Build a dashboard (Streamlit / Power BI / Web-based interface).
- Display trends, alerts, and activity heatmaps.

6️⃣ Expected Output

- A trained AI model for satellite-based activity detection.
- A structured database of detected logistics activity.
- A dashboard displaying monthly trends and alerts.
- A proof-of-concept early warning system for supply chain disruptions.

7️⃣ Business Value

- This project bridges:
- Remote Sensing
- Artificial Intelligence
- Business Intelligence
- Operational & Financial Analysis

It demonstrates how satellite-based monitoring can anticipate supply chain bottlenecks, reduce operational risk, and support strategic decision-making.

8️⃣ Innovation Aspect

Unlike traditional academic projects, this system integrates:

- Real-world business KPIs
- AI-based satellite analysis
- Predictive operational monitoring

This makes the project highly applicable in logistics companies, investment firms, and supply chain analytics platforms.

9️⃣ **Mathematical Modeling Framework**

To strengthen the analytical foundation of the project, we propose a mathematical framework that formalizes satellite-based activity monitoring into measurable indices and predictive models.

9.1 Object Detection as an Optimization Problem

Let an input satellite image be represented as:

$$I \in \mathbb{R}^{H \times W \times C}$$

where:

- H= height
- W= width
- C= number of channels
- The object detection model learns a function:
- $f_\theta(I) = \{(b_i, c_i, s_i)\}_{i=1}^{N}$

- where:
- b_i= bounding box
- c_i= object class (ship, truck, container, etc.)
- s_i= confidence score
- Training is formulated as minimizing a multi-task loss:
- $\mathcal{L} = \mathcal{L}_{classification} + \mathcal{L}_{localization}$

- This corresponds to a constrained optimization problem:
- $\theta^* = \arg\min_\theta \mathcal{L}(\theta)$

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- **9.2 Activity Index Modeling**
- Let:
- S_t= number of ships detected at time t
- T_t= number of trucks

- C_t= number of containers
- We define a generalized activity index:
- $A_t = \alpha S_t + \beta T_t + \gamma C_t$

- where:
- $\alpha, \beta, \gamma$are weighting coefficients
- Weights can be estimated using regression or normalization techniques
- Example: **Port Congestion Index**
- $PCI_t = \frac{S_t}{\bar{S}}$

- where $\bar{S}$is the historical average.
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- **9.3 Time Series Modeling**
- We treat activity as a stochastic process:
- $\{A_t\}_{t=1}^{T}$

- Possible models:
- ARIMA model
- Exponential smoothing
- LSTM neural networks
- General autoregressive representation:
- $A_t = \phi_1 A_{t-1} + \phi_2 A_{t-2} + \cdots + \epsilon_t$

- This allows:
- Trend detection
- Seasonality modeling
- Anomaly detection

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- **9.4 Change Detection as a Statistical Test**
- Let two images at times $t_1$and t_2:
- I_{t_1}, I_{t_2}

- We define a difference function:
- $D = \| f_\theta(I_{t_1}) - f_\theta(I_{t_2}) \|$

- We can model change detection as a hypothesis test:
- H_0:No significant change
  H_1:Significant activity change

- If:
- $D > \tau$

- then we reject H_0.
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- **9.5 Risk Score Modeling**
- Route risk score can be defined as:
- $R_t = w_1 C_t + w_2 D_t + w_3 F_t$

- where:
- C_t= congestion measure
- D_t= detected damage
- F_t= flood probability
- w_i= calibrated weights
- This can be treated as a weighted linear model or optimized via:

- $\min_{w} \sum (R_t - y_t)^2$

- if labeled risk data is available.

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- **1 0 Mathematical Contribution of the Project**
- This project integrates:
- Optimization theory
- Statistical inference
- Time series analysis
- Weighted index modeling
- Hypothesis testing
- Applied stochastic processes
- It transforms raw satellite imagery into mathematically defined operational indices, making it not only an AI project but also a quantitative modeling framework.